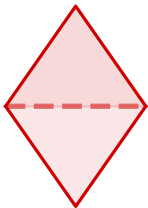


(a)

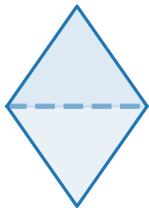
Semenoff (inversion) mass

$$C_- = 0 \text{ (cancel)}$$

 $K (\tau = +1)$ 

$$\Delta_K = +m_S$$

$$C_-^K = -\frac{1}{2} \text{sgn}(m_S)$$

 $K' (\tau = -1)$ 

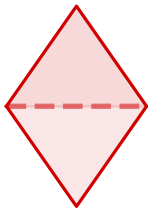
$$\Delta_{K'} = +m_S$$

$$C_-^{K'} = +\frac{1}{2} \text{sgn}(m_S)$$

(b)

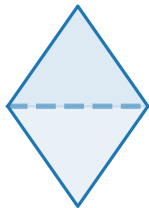
Haldane (Faraday) mass

$$C_- = -\text{sgn}(m_H) \text{ (add)}$$

 $K (\tau = +1)$ 

$$\Delta_K = +m_H$$

$$C_-^K = -\frac{\text{sgn}(\Delta_K)}{2}$$

 $K' (\tau = -1)$ 

$$\Delta_{K'} = -m_H$$

$$C_-^{K'} = +\frac{\text{sgn}(\Delta_{K'})}{2}$$

Convention: $H_\tau(\delta \mathbf{k}) = v_D(\tau \delta k_x \sigma_x + \delta k_y \sigma_y) + \Delta_\tau \sigma_z$; $C_-^K = -\text{sgn}(\Delta_K)/2$, $C_-^{K'} = +\text{sgn}(\Delta_{K'})/2$